

Living Alignment

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Abstract

There is broad agreement that the goal of AI alignment is to promote futures full of health and flourishing, but our understanding of what that means and how to reach it remains poor. Here we look to the science of life for guidance. Across scales, from genomes to ecosystems, we observe that notions of health and flourishing consistently map to the ways in which those systems limit fixation on any particular mode of interaction. Fixation is limited by barriers that contextualize modes of interaction, leading to robust modularity and open-ended invention of new structure. With these principles in mind, we turn to the alignment problem. We suggest that to be aligned in a meaningful sense, AI and human-AI systems must limit fixation and include evolving barriers at all scales.

Keywords: AI alignment | AI safety | evolutionary biology | open-endedness | value specification | sociotechnical alignment | human–AI systems

‘Defying definition—a word that means “to fix or mark the limits of”—living cells move and expand incessantly.’

Lynn Margulis

‘This web of life... breaks no law of physics, yet is partially lawless, ceaselessly creative.’

Stuart Kauffman

In this Perspective, we draw on the science of living systems to better understand how to design and relate to AI in ways that lead to futures full of flourishing and health. Understanding what makes systems healthy in a very general sense might offer some hope of reasoning about futures that could evolve rapidly as AI develops.

In Section 1, we point to a principle of life which has been noted across a variety of systems at different scales. The principle is that living systems are healthy and flourishing when they avoid overly fixating on, or reducing to, particular patterns of interaction. Consistently across types of system, fixation is limited by semi-permeable barriers, which contextualize patterns without negating them. Barriers translate shorter-sighted optimization pressure into productive effort within more flexible structures. This process often yields deepened individuality, modularity, and open-ended invention. We work through examples of these mechanics at scales from cells to societies.

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Section 2 applies the same principle to the AI alignment problem. There is broad agreement that the aim of alignment—in its broadest sense—is futures full of health and flourishing. From the template identified in Section 1, it follows that alignment means avoiding fixation on particular modes of interaction. This simple lemma has a surprising consequence: any alignment method, if given too much weight, is necessarily *misaligned*. Instead, the flourishing of life in the future depends on an evolving set of barriers that continue to limit the influence of any particular approach or concept, creating a generative web of partially competing, partially collaborating perspectives at all scales. Ongoing invention is intrinsic to alignment.

1 Health in living systems

Section 1 works through examples in a range of living systems. Across the example systems, health and flourishing are related to limiting fixation on particular modes of interaction. By limiting fixation, healthy living systems sustain a buoyant internal structure that admits many kinds of description but is not perfectly or eternally captured by any of them. Conversely, systems are unhealthy when some mode of interaction gains excess traction or permanence at the expense of broader diversity, flexibility and dynamism.

Fixation is limited in successful living systems by barriers or limits on particular modes of interaction. Some barriers are features of the non-biological world: the speed of light, rates of reaction, mountain ranges. Life also places limits on itself. Humans precommit to limit their own future impulsivity, plants deploy elaborate barriers against self-fertilization, and a gene’s selfishness is held in check by other parts of the genome. A barrier is a mechanism to limit absolutism.

Effective barriers are semi-permeable, permitting interactions that are temporarily or conditionally stable, without letting them become rigid or dominating. The conditionality is often highly specific. Barriers might block certain objects or information, constrain when interactions happen or how large an effect they may have, or gate information depending on another factor. The conditions are sometimes fine-tuned through the history of the system, with each part carrying traces of the contexts it has participated in. Conditionality contextualizes a local perspective as part of a larger system.

Limiting fixation on simpler modes of interaction makes living systems generative. Without barriers, systems tend to stagnate as parts become overfit to particular configurations and optimization pressures are satisfied through degenerate solutions. With barriers, the system rides a delicate edge where each entity tends to deepen its unique identity, establishing a modular basis that becomes the building blocks of invention. As a result, more sophisticated structures can form that wouldn’t have been possible without the diverse parts each having undergone their own unique development. The larger system accommodates many partial perspectives rather than locking in a single story of what matters, and systems are elements in other systems, creating a deep heterarchy. At the sensitive edge of semi-stability, life open-endedly continues to generate new kinds of organization that require fundamentally new descriptions (Canguilhem, 1966; Kauffman, 1992; Langton, 1990; Stanley et al., 2017).

Institutions

Every actor in a society or an economy is myopic in some way, valuing its own welfare over others’, discounting the future, or attached to a particular perspective. If unchecked, an individual actor’s optimization for its own agenda can be harmful: a company’s profit motive leading to suppression of competition, deception and exploitation (Bakan, 2006); desire for dominance leading to disempowering and silencing others (Sidanius and Pratto, 2001); even prosocial beliefs producing conflict and suppression when groups see differently (Haidt, 2012). Each of these is an example of fixation, where the myopic actor gains too much traction and collapses some of the nuance of the larger system.

Institutions—constitutions, laws, norms, courts, regulators—are barriers against domination by any particular actor’s motives (Leibo et al., 2025; North, 1990; Ostrom, 1990). The point of institutions is not to annul the short-sighted intentions of individual actors but rather to contextualize them. When functioning well, institutions productively reroute motivations: for example, a business’s profit motive, when constrained by competition and effective regulation, may fuel innovation. Since intelligent agents

do not necessarily accept the barriers set on their desires, institutions must adapt as loopholes are found, forming an evolving network.

Ecosystems

Ecosystems similarly contain an evolving network of constraints, such as predation, parasitism and resource competition. These interactions support the richness of the larger system. In Paine’s classic experiment, removing predatory sea stars from a rocky shore allowed mussels to expand and displace other organisms competing for space (Paine, 1966). Without predation as a barrier, excess influence of mussels impoverished the local ecosystem.

While lack of barrier can lead to collapse, presence of barriers ideally sets up an intricate balance in which new niches and new functions can emerge which previously did not exist. Beavers, for example, alter water flow and create wetland habitats, increasing plant richness across the landscape (Wright et al., 2002). The network of constraints translates local optimization pressure into generative potential.

Of course, health and fixation are always relative. Extinctions can open opportunities for innovation; the death of a cell is catastrophic for the cell but may be necessary for the organism. Life’s history is littered with local fixations at all scales. When they are contextualized, the larger system can remain healthy.

Groups

Teams, companies, and civilizations achieve what no individual can (Hutchins, 1995; Woolley et al., 2010). The greatest achievements depend on structured interactions of individuals who have developed their own specializations and perspectives.

Conversely, group performance can degrade when groups homogenize (Bernstein et al., 2018; Frey and Van de Rijt, 2021; Hong and Page, 2004; Janis, 1972; Lazer and Friedman, 2007; Lorenz et al., 2011). Zollman (2010) tells the remarkable story of how a single study failed to find bacteria in the stomach of peptic ulcer patients. The study was too influential: the belief that ulcers are not caused by bacteria propagated through the field, and medicine settled on a wrong understanding for decades.

Barriers encourage diversity and generativity. For example, Bernstein et al. (2018) tasked small groups with solving traveling-salesman problems. In these problems, it’s possible to get stuck in dead-end solutions that initially appear promising. In Bernstein et al.’s experiment, groups performed better when they were not allowed to exchange information continuously. Working independently, group members pursued their own diverse solutions, some of which turned out to be superior—rather than picking up on seductive dead-ends from others. The barrier was semi-permeable in the sense that isolation was temporary, followed by interchange.

In the real world, aggregation is often modular and compositional. The steel punch was developed by goldsmiths to imprint many coins with the same design. Independently, farmers had spent centuries optimizing the screw press to extract oils and juices. Gutenberg drew on both technologies to construct his printing press (Johnson, 2010). A group with many ideas and specializations available, each developed through the unique pressures of semi-isolation, is ripe for such compositional discoveries (Surowiecki, 2005).

Frames

Humans always look at the world from some perspective; there is no ‘view from nowhere’ (Nagel, 1986). Most real situations admit many valid perspectives, and each is only a partial description—famously, ‘all models are wrong’. But for an agent inside a frame, the frame is invisible; its organizing assumptions are experienced as features of reality itself (Berger and Luckmann, 1966; Goffman, 1974; Wittgenstein, 1953).

Losing the ability to move between frames is a signature of psychopathology (Kashdan and Rottenberg, 2010). In depression and anxiety, inflexibility manifests as rigid and repetitive negative beliefs about self, the world, and the future (Beck et al., 2011). In obsessional disorders, people report distressing (ego-dystonic) intrusive thoughts, which, although recognised as incorrect and maladaptive, are nevertheless hard to resist. People experiencing schizophreniform or affective psychoses may experience bizarre delusions that are at odds with available evidence and cultural norms, yet are held with conviction and resistant to evidential challenge (Adams et al., 2013).

A frame is invisible from inside it, but a nifty capability of humans is stepping out of their frame—reorienting to see it as an object. Something that previously was unconsciously axiomatic becomes instead a content of awareness, a process variously referred to as decentering, defusion, disidentification or dereification (Bernstein et al., 2015; Kegan, 1982; Lutz et al., 2015). Decentering places a critical barrier limiting fixation on any particular belief or mental form. The barrier is semi-permeable: becoming aware of a belief does not make it wrong in an absolute sense any more than it was right in an absolute sense, and it is held productively for the partial truth it contains. At the delicate edge of less exclusive fixation on particular forms, subtler forms, which could have been erased by clamped fixation, instead play a role in a rich internal structure.

The barrier of decentering is generative. An unquestioned belief sets up a fixated mode of interaction: the belief has a ‘free pass’ and is doggedly applied. When a belief no longer has this protected status, it is tested against evidence and against other beliefs. This forces the belief to become a better model of the world, creating a deeper repertoire.

Goals

Animals have many innate drives: for nutrition, osmotic balance, temperature, reproduction, avoidance of pain. Each evolved as a proxy for fitness, but each is imperfect, so fixation on any one paradoxically reduces fitness rather than increasing it (John et al., 2023; McFarland and Sibly, 1975). Maximizing caloric intake produces obesity; single-minded pursuit of sex produces relational, occupational and legal harm. Higher-order goals behave the same way. Fixating on work goals can cause burnout, and maximizing reported revenue without regard for honesty or law can lead to imprisonment.

Each goal is partial—it only makes sense in some contexts. For example, in a classic experiment, hungry chickens were placed near a cup of food rigged to move in the same direction as the chicken at twice the speed, so that food could only be obtained by running away from it. The chickens’ fixation on the prepotent, overly simplistic aim of running physically toward the food led to a worse overall outcome (Dayan et al., 2006; Hershberger, 1986). The chickens could not contextualize the ‘approach food’ strategy to use it only when appropriate for the environment’s structure.

Evolution, as well as learning and culture, provide an array of barriers to limit fixation on particular drives and goals. Some barriers are homeostatic, where a drive is downregulated when it is less necessary. Another broad class of barrier is cognitive control (Diamond, 2013; Miller and Cohen, 2001). Yet another class of internally-generated barrier is precommitment devices, where a person contrives that their future self is less likely to suffer from fixated goals. These barriers are often semi-permeable. For example, cognitive control facilitates contextual switching between motives but does not nullify them.

Placing constraints on fulfilment of shallow objectives sometimes leads to progress on deeper ones (Duncker, 1945; Ohlsson, 1992; Rosenberg, 1969; Stokes, 2005). Animals sometimes develop novel solutions to problems when their default approach fails to produce rewarding outcomes. When people voluntarily give up access to Facebook, they spend more time socializing offline with friends and family (Allcott et al., 2020).

Brains

The brain keeps many pieces of information distinct from one another while allowing them to interact. This is a distinguished achievement in highly-connected networks. In fact, we speculate that the mechanisms of life’s generativity are most finely honed in the brain.

Inhibitory circuits constrain the strength and timing of excitation and sharpen stimulus selectivity (Isaacson and Scanziani, 2011). Popular models of neocortex describe descending signals as canceling out any predicted activity, so the activity that propagates is specifically what is orthogonal to predictions (Rao and Ballard, 1999). The circuit architecture of the hippocampus creates sharp representational distinction between different experiences, preventing interference between memories and concepts (Leutgeb et al., 2007; Yassa and Stark, 2011).

Compared to other animals, the human brain especially attempts to discretize its experience into approximately symbolic representations (Behrens et al., 2018; Dehaene et al., 2022; Smolensky, 1990;

Touretzky and Hinton, 1988). Separating knowledge into distinct entities and then recombining them in vast numbers of structured ways is believed to power the extraordinary human capacity for reasoning and generativity (Chomsky, 1957; Fodor, 1975; Kurth-Nelson et al., 2023; Lake et al., 2015; Pinker, 1994; Schacter et al., 2007).

More broadly, healthy brain dynamics live at a sweet spot between excessively stable synchronized patterns and chaotic uncorrelated noise (Chialvo, 2010; Deco et al., 2011; Haldeman and Beggs, 2005; Rabinovich et al., 2008; Shew et al., 2011; Tognoli and Kelso, 2014). In this regime, the brain has access to a huge repertoire of patterns it can explore temporarily without overcommitting or getting stuck. Loss of dynamic flexibility, where the brain’s activity becomes more stereotyped and no longer explores as wide a repertoire of states, is tied to lower cognitive performance (Cocchi et al., 2017; Grady and Garrett, 2014; Müller et al., 2025). More extreme stereotypy corresponds to severe dysfunction. For example, in Parkinson’s disease, basal ganglia and cortical circuits collapse into excess synchrony and lose the flexibility needed to guide nuanced motor outputs (Brown, 2003; Hammond et al., 2007).

Cells

The endosymbiotic theory explains the origin of mitochondria and chloroplasts as ancestral bacteria that got incorporated into host cells and evolved into organelles (Sagan, 1967; Zimorski et al., 2014). In this theory, the predecessors of the incorporated cells evolved for millions of years as lineages distinct from the predecessors of the host cells. Through this partial separation, they developed distinct specializations. Symbiogenesis could then combine capacities developed separately into a new larger structure which may have been difficult to discover directly through evolution on a single type of organism (Watson and Pollack, 2003).

The sheer fact of maintaining distinct evolutionary experiments over a great length of time—the predecessors of the host cells and the predecessors of the organelles—is sometimes underappreciated. If those lineages had blended into a single type, it is plausible that the components would not have existed from which to build higher-order structure.

Genomes

Sex is costly. An allogamous organism must find a mate in a vast and dangerous world. In species with separate sexes, roughly half of the population does not bear offspring (Maynard Smith, 1971). And meiosis causes each parent to lose half its genetic representation in each child (Williams, 1975). Yet nearly all eukaryotes reproduce sexually (Speijer et al., 2015), and many hermaphrodites go to great lengths to prevent self-fertilization (Charlesworth et al., 2005). Why place a barrier on the path to reproduction?

In an asexual lineage, genes are permanently locked together: selection cannot act on one without dragging the others. A deleterious mutation is inherited forever, unless the lineage is lucky enough to experience the reverse mutation (Muller, 1964), and two beneficial mutations arising in different individuals cannot be subsequently combined into a single genome (Fisher, 1930; Muller, 1932). Asexual reproduction or self-fertilization creates a high degree of informational connectedness between the genes of a genome. Every gene has a ‘free pass’ into future generations within a lineage, and alleles become overfit to a fixed genetic background.

Recombination breaks up associations among loci, allowing selection to act more independently on them (Felsenstein, 1974). By exposing alleles to varying genetic backgrounds, it favors alleles that perform well across contexts, promoting modularity and evolvability (Livnat et al., 2008, 2010; Wagner and Altenberg, 1996; Watson and Szathmáry, 2016). The result is modularity, where each gene becomes an interchangeable unit that can contribute something meaningful to many different genetic contexts. This creates the foundation for faster discovery of new solutions.

Geographic and reproductive isolation are another kind of barrier, which allows populations to accumulate distinct variants. Subsequent introgression brings these variants into new combinations and can contribute to rapid innovation (Marques et al., 2019).

Summary

We still can't reproduce life. We don't understand very basic things about how life started or whether drastically different forms of life could exist without Earth-like chemistry.

Life has developed (both at evolutionary and cultural timescales) through immense amounts of feedback from the real world, which shaped it at many levels.

The picture of life as autonomously fighting on every scale. The matroshka doll.

2 The alignment problem

In Section 1, we observed that living systems are healthy when they avoid fixation, instead holding the delicate, generative interplay of many partial perspectives. This interplay breaks down when any particular pattern of interaction gains too much traction, suppressing nuance and reducing creative potential. Semi-permeable barriers both promote the individuality and autonomy of distinct entities and also enable relational participation, holding local perspectives strongly enough to act but lightly enough to remain open to broader context. Under such conditions, systems open-endedly explore fundamentally new forms, avoiding collapse to static equilibrium or fixed compromise.

Now we look at the alignment problem through this lens. There is broad consensus that alignment means working toward healthy and flourishing futures. We therefore reason that alignment can be understood as the problem of maintaining creative sensitivity by limiting fixation on any particular form.

An advantage of this perspective is that it is grounded in principles that underpin life at a very general level. We accept that humans, other living systems and AI are tightly interwoven (Bateson, 1972; Bijker et al., 2012; Folke, 2006; Haraway, 2016; Hayles, 1999; Latour, 1993); moreover, rapid advances in AI may lead to large changes in the organization of the world. A systems analysis is both useful for thinking about flourishing in the present and more likely to generalize to substantially different futures.

No scheme is alignment

One implication of this perspective is that any alignment scheme must be incomplete. Any form, pattern or conceptual scheme is misaligned if taken too seriously. Therefore no particular approach can achieve alignment, and continuing alignment intrinsically depends on continuing invention. An AI system could fixate on the language for describing the space that goals and values live in; an algorithm for learning human preferences; methods for reaching consensus; a constitution; concepts of what value, agency, uncertainty or representation mean; or foundational aspects of our beliefs about the world that we take for granted but have difficulty seeing. Even an idealized or inferred set of values, such as coherent extrapolated volition (Yudkowsky, 2004), is misaligned to the extent that it has a fixed form and that fixed form is overly relied on. This is the lesson of life: healthy living systems are endlessly generative, with autonomous perspectives loosely and modularly coupled at many scales.

Importantly, this does not mean we should have no alignment schemes. Quite the opposite: they are critical barriers. Discarding schemes capriciously can be as much a fixation as attaching to them doggedly. Continuing to discover new schemes and formalizations is essential to making progress.

Reward hacking

To give a flavor of the argument, we work through one example of a problem in AI safety, a method used to tackle that problem, and why the method can't be a final answer.

Reward hacking is the problem that an AI system, given some definition of what is good, finds loopholes in the definition. Exploiting these loopholes, the system produces outputs that are good according to the definition but clearly undesirable. For example, Anthropic reported that LLM-based agents, which were scored on their task outputs being equal to correct outputs, avoided solving the task by programmatically redefining the 'equals' operator to always return 'true' (MacDiarmid et al., 2025). Analogously, AI systems rewarded for positive human judgements sometimes become flattering and sycophantic (Sharma et al., 2023). The stronger the AI system is, the more potential there is for exploiting loopholes to have catastrophic side effects—for example, a superpowerful AI tasked with maximizing human happiness might

implant electrodes in our brains to stimulate pleasure (Bostrom, 2014). A reward hacking agent fixates on a narrow definition of values, collapsing the broader context.

Reward hacking is challenging to avoid because our true values are incredibly complex. Humans have many kinds of values (Nagel, 1979). We value the short-term hedonic impact of sugar on the tongue; we also value our long-term health, the welfare of our grandchildren, the wellbeing of strangers in another country, even the sustainability of the planet. We value different things at different times, and different people have different values (Leibo et al., 2026; Rawls, 1993). Given time and resources to deliberate, we may attest to different values than when we make snap decisions (Suter and Hertwig, 2011; Yudkowsky, 2004). Some of our values are difficult to express verbally, stretching below language into subtle, contextual intuition involving our bodies, communities and natural environment (Abram, 1996; Johnson, 2014; Polanyi, 1966; Varela, 1999). We may want to leave room for new values that emerge in the future, which could conceivably be as different from ours as ours are from a trilobite’s. With such complexity and non-stationarity, it is almost inevitable that however we formally describe our values, the description will leave loopholes.

Numerous methods have been developed to mitigate the problem of reward hacking. One interesting class of methods treats values as intrinsically uncertain and therefore something to be learned in an ongoing way from the behavior of humans or other indirect signals (Hadfield-Menell et al., 2016, 2017; Jeon et al., 2020; Russell, 2019; Shah et al., 2020). An agent unsure of what humans want should become more risk-averse and open to correction. For example, a human’s attempt to turn it off can be construed as information about human preferences. Such a scheme is a powerful barrier limiting fixation on one reward function. However, it is necessarily incomplete. Any particular method for learning values is still fixated if there are aspects of it that are constant across time as AI systems become more powerful and the world changes. Rather than being fixated on a particular value function, the system may be fixated on something more abstract: what counts as human behavior, what counts as a human, how to map behavior to values, the formalism for describing values, how to represent and update uncertainty, how value is translated into action. If any of these formalisms are fixed while the systems built around them become increasingly powerful, the outcomes are likely to be incompatible with open-ended flourishing of life.

Is AI special?

So we’re saying that we need to keep developing new barriers in an ongoing way. Isn’t that what humans have always done?

With AI, things change faster, and there is a lot of power behind fixed forms. In the past, there have always been arms races between different patterns of organization, and sometimes one gets so far ahead that the local area is decimated. But if a version of AI gets so far ahead of everything else, there is the possibility of world-level destruction. AI systems increasingly are acting autonomously, faster than humans can follow, in ways humans cannot understand.

Why should AI be different from past technologies? We do not expect what we build today to be perfect; we expect to keep finding and fixing problems, which is what humans have always done. However, the iterating process may not continue to work as well as it has in the past. AI already has remarkable properties: rapid global adoption, intensive use of resources, concentration of information flow, anthropomorphization by humans, offloading of much cognitive activity, and rapid improvement. Conceivably, future AI may exhibit superhuman intelligence and recursive self-improvement without being limited to a restrictive biological substrate. Compared with past technologies, these properties may make it harder to iterate on imperfect solutions (Bostrom, 2014), and any particular forms or assumptions built into AI systems could have substantial consequences.

Example: concentration of power. Humans have different interests, and AI is likely to be aligned to the interests of some more than others (Gabriel and Keeling, 2025; Sorensen et al., 2024). Advanced AI might also confer substantial power on those who control it, raising the possibility that a small number of humans use that control to dominate others—more likely as persuasive power increases (Costello et al., 2024), autonomous weapons place lethal force in few hands (Scharre, 2018), surveillance improves, and the need for human labor decreases (Susskind, 2020). In either case—AI’s unfairness or human power consolidation—extreme inequality risks collapsing society to the goals of a few individuals. Of course,

a world where everyone is identical would be another form of fixation; distributing resources and power mitigates against fixation to the extent that it enables many diverse identities to thrive at multiple scales. It could also go the other way: because roughly the same AI technology is available to the poor and unskilled as to the wealthy and skilled, some have proposed that AI will have a levelling effect (Brynjolfsson et al., 2025).

Example: conceptual monoculture. Conceptual monoculture is fixation on particular beliefs, ideas, frames, values or problem-solving approaches—loss of diversity across a group. Precious diversity exists at every level: within one person, between humans, between humans and AI systems, and between organizations, fields, cultures and nations. Its collapse threatens fragility and lower performance (Page, 2008). Current AI systems draw from a conceptual manifold that is, in some ways, impoverished relative to humans (Messerli and Crockett, 2024), and recent studies find that while individual AI outputs may be judged superior to human ones, they are also more homogeneous (Doshi and Hauser, 2024; Padmakumar and He, 2023). This narrow manifold might get broadcast to the whole world: at least a billion people now use AI for everything from relationship advice to industrial maintenance (Chatterji et al., 2025), yet because frontier models are difficult and expensive to produce, that usage is routed through a handful of models (Bommasani, 2021), risking global loss of diversity (Sourati et al., 2026). And because humans are influenced by AI while human outputs are training data for future models, homogenization could become recursive (Chaney et al., 2018). It is worth noting, however, that none of these studies made a best-effort attempt to use AI thoughtfully to increase rather than decrease diversity; with exposure to vast amounts of data, AI has the potential to retain and even create many novel distinct perspectives.

From alignment to living alignment

So what do we do,

It follows that it is not enough to simply ‘do what’s good for humans’ in a naive sense. That notion breaks down. Humanity and AI may both change substantially, and future worlds may contain patterns of organization that do not exist yet—perhaps some that would be incomprehensible to us at present. Things will change, including what there is to value and how it is valued. Think of the world today compared to the world a billion years ago. Any formalism or concept we have now will eventually appear as limited as a single-celled organism’s notions of ‘what is good and bad’.

Therefore, the problem of alignment, if carefully inspected, transforms into the open-ended problem of living alignment. In order to be aligned in the long run, AI must become more lifelike. Otherwise, as it has more power, that power will fixate particular forms. Living alignment means life-like systems that open-endedly discover new ways of contextualizing any myopic processes, preserving and extending the richness and generativity of life. As human-AI systems develop, they will stay aligned if they continue the journey of life by building semi-permeable barriers to contextualize particular forms and nurture existing and unfolding subtlety at many scales.

end on healthy outcomes for transition...

Healthy for whom?

Our framework defines aligned systems as those with lifelike properties of multi-scale agentic modularity, which avoid collapse onto fixated forms. Is this ‘alignment’? Is it alignment to ‘human values’? How can you align to something if you start with the assumption that that thing is going to fundamentally change? Alignment is typically concerned with human interests. What happens if AI replaces humans with its own lifelike systems, but destroys humanity in the process?

As we’ve discussed, fixation and health are always relative. Local fixation is part of larger flourishing, and of course the interests of different entities sometimes trade off against each other.

How is the flourishing of living systems related to human values? Our suggestion is that the living process is both inclusive of existing human values and aligned with continued open-ended flourishing in futures where qualitatively new values may emerge. Eventually, if there are descendants of humans, they will presumably be drastically different from what we look like at present. Holding things in stasis eternally is not what we want. Even though current human values are unlikely to be the ultimate endpoint of the

universe, they are an important part of its tapestry and depth.

The life that exists today, now including humanity’s storage and processing of information, is what traced a path through time avoiding both excess and insufficient stability (Maturana and Varela, 2012; Prigogine and Stengers, 1984; Schrödinger, 1944).

However, in the short to medium term, honoring existing life and values is where it’s at. It is difficult to overstate the amount of complexity and content in existing life, including in the relationships between organisms and in the mental lives of humans. Existing life has been through so much to get here: billions of years of parallel experiments all over Earth. Living alignment is inclusive of this diversity of evolving values. We saw consistently in Section 1 that life gives separate entities space to develop distinctly without being overwritten or homogenized, and that these parts never-endingly join in structures that sustain their individuality while situating them as only part of a larger story. Another way of saying this is that life steps back to take into account more of the context. AI comes into existence amid a profound network of existing reality saturated with meaning and importance, and the diversity of human—and perhaps non-human—values is part of that context. Rather than short-circuiting into fixated modes, an aligned future continues to expand toward solutions inclusive of more and more apparent contradiction.

Going forward

The industry is already building in internal barriers. We apply safety post-training, guardrail models, red teaming and ongoing evaluation, and we try to improve our mechanistic understanding of AI systems in order to detect and correct hidden flaws (Olah et al., 2020). OpenAI now continuously monitors the activations and thinking traces of models, to detect when internal activity becomes suspicious, for example if models are attempting to circumvent safeguards (OpenAI, 2026). Fair principles have been proposed to prevent over-reach of any party (Gabriel and Keeling, 2025), and governments apply oversight to anticipate and mitigate risks to human rights and national security (European Parliament and Council of the European Union, 2024). Concentration of power might be mitigated by democratic oversight and involvement of more people in AI design decisions (Birhane et al., 2022), or by redistribution mechanisms (O’Keefe et al., 2020). OpenAI and Anthropic have paused development or release of frontier models to strengthen safety mechanisms (Anthropic, 2026; OpenAI, 2026). Some of these barriers have humans in the loop to greater or lesser degrees, like governmental oversight, mechanistic interpretability and scalable oversight. Others are fully automated, like automated chain-of-thought monitoring, agentic red teaming and guardrail models.

An aligned future rests on the continued generation of, and sensitivity to, new forms and concepts, including new values.

An aligned future has the generativity, robustness and modularity of life at all scales.

Again, any fixed conceptual approach is misaligned. The essence of living alignment is transitioning to a system—which includes humans, AI, and the other species and ecosystems on Earth—that carries forward the full momentum and generativity of life, rather than fixating. As AI develops new capabilities, contributes to its own development, and transforms more aspects of human life, alignment will require an increasingly elaborated array of semi-permeable barriers, because the potential modes of collapse will keep changing as the world changes. We do not know how likely it is that appropriate barriers will emerge. As in triage, there are three possibilities: myopic forms will inevitably be supercharged by progress in AI beyond the ability of anything else to bound them, resulting in disaster; or appropriate barriers will inevitably emerge to contextualize any forms of AI; or the outcome depends on the choices we make now.

The coarsest versions of fixation are obvious: paperclips, bioweapons, extreme concentration of power. What we have to be careful about are less obvious versions. A subtle fixation at first appears harmless, but the system as a whole is collapsing toward something without contextualization. The pessimistic argument is that AI is eroding barriers beyond a healthy balance. AI systems could give great weight to narrow goals or ideas. They might facilitate rogue actors building weapons of mass destruction, or reduce the ability of democratic systems to check the power of a few individuals or authoritarian states. They might give everyone similar information and damage conceptual and cultural diversity. They might execute their own narrow goals destructively. If people and information systems across the globe become hyper-correlated,

it is as if there were a single superorganism on Earth, which may reduce both resilience to shocks and the potential for future evolution—and past evolution always depended on running parallel experiments with diverse and separate organisms. As we reviewed with problem solving in groups, there is a simple concept in modeling studies that excessive network connectivity is prone to reaching mediocre solutions quickly and losing the diversity that sustains potential for reaching higher quality solutions over the long run (Lazer and Friedman, 2007; Zollman, 2010). [Conceptual monoculture leads to an impoverished future, or a loss of drive.](#)

The optimistic argument is that AI is creating more and more structure in the world, including the appropriate barriers [to contextualize its own new abilities](#). We have AI tools for bounding AI: detecting bugs before others can exploit them, detecting deepfakes, scalable oversight. Appropriately-tuned AI can facilitate human learning and nurture each individual’s unique perspective and curiosity, and democratic AI could help humans with different perspectives co-exist and cooperate. Advances in technology may create new niches faster than they destroy old ones, and there is plenty of space on Earth for increasing diversity.

[The good version increasingly looks like subject-to-object perspective-taking within AI and human-AI systems.](#) At the sensitive edge between excess belief and excess disbelief, we sometimes, individually or collectively, step back to hold a previously axiomatic thing in a larger context. This kind of discovery—transformation of the subject itself—is arguably a central aspect of our own livingness and of the subtle livingness on the planet. As AI systems gain agency and ability and become integral parts of social and psychological systems, internal discovery may also become an essential feature of AI and human-AI systems. We imagine living AI systems continually contextualizing their own foundational assumptions as partial truths, participating in the open-ended process of life. [This is internal discovery within systems that represent themselves and the world.](#)

The melioristic argument is that because it is easy to imagine concrete paths toward both outcomes, both may be within the realm of possibility; and that as long as there is any possibility our choices are decisive, we are obliged to act as if they were (Ord, 2020). We lean toward the melioristic position. If it is right, then continual effort toward invention will be required—first from humans, and then increasingly from AI or human-AI systems—to contextualize myopic forces. Many of the ways this effort would be applied are already well studied in AI safety. Others will have to be discovered. No set of barriers will be a final answer.

[The ideas in this paper come with the same proviso: we should not over-attach to them.](#) Our aim is to offer a conceptual lens, which is of course imperfect and temporary.

Conclusion

We learn from living systems that health and flourishing are not any particular form or concept. Therefore alignment is not picking the right values or principles, or even the right system for learning them. It is not any method for interpretability or corrigibility. All of these can be useful parts of alignment.

[Instead](#), an aligned future, through internal and external discovery, continues the living process of maintaining and enhancing the individuality of distinct elements while deepening their relatedness—like how a species over time acquires more distinctive features while also becoming a richer reflection of the world around it. Distinct elements continue to experiment with their own solutions to the problems they face from their unique perspectives, expanding a broad reservoir of uniquely useful components that can be drawn on by larger systems that include them. Every entity is always inevitably myopic, but the world as a whole continues to unfold in a robust way because different perspectives recast each other’s sharp edges as constructive momentum.

[Alignment itself](#) is the continued dance of contextualizing any particular form, more fully respecting the staggering subtlety of the existing world, and open-endedly supporting new robust depth that admits more and more different kinds of metaphor or description. In this way, AI can participate in intense flourishing of an evolving world even far beyond current human conceptualization. The universe thrives on an open-ended trajectory of diversity, subtlety and potential.

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Example	Fixation	Barrier	Generativity
Symbiogenesis	Loss of differentiated ecological and genetic capacities (hypothetical)	Distinct niches and limits on gene exchange; subsequent organelle compartmentalization	Integration of capacities developed in distinct lineages into the eukaryotic cell
Sexual reproduction	Linkage disequilibrium, Muller’s ratchet	Sexual reproduction, out-crossing	Each gene explores a solution that can contribute to many genomes, creating combinatorial possibilities
Biological neural networks	Hypersynchrony (depression?)	Homeostatic plasticity, recurrent inhibition, etc (could also do small-world with local specialization)	Behavioral flexibility
Pavlovian approach	Persisting in approach even when it doesn’t work	Arbitration, cognitive control	Exploring, building a richer model
Frames	Inflexible beliefs, obsession, functional fixedness	Transition to perspective-as-object	New, more inclusive perspectives
Group problem solving	One perspective dominates, group performs worse	Structured communication, skepticism	Combining unique ideas create new inventions (eg printing press)
Institutions	Theft, monopoly, resource overuse, concentration of power	Laws, regulations, courts, etc	Individual motives become productive parts of systems
Ecosystems	Collapse of biodiversity	Predation, competition, etc	Niche creation

Table 1: Examples of fixation.

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